

Cohesion, Matrix Tension, and Impact-Site Sensitivities in DART-RIFT Simulations of Dimorphos

Yağız Ekrem Dalar

ekrem@ethosoft.org  0009-0009-3522-4403

Feyzi Arda Salihoğlu

feyziarda.salihoglu@ethosoft.org

 0009-0006-5841-1664

Nedim Mutlu Sezer

nedim@ethosoft.org

 0009-0001-3499-6067

GO İstanbul Girişim Ofisi (Bilişim Vadisi), Sarıyer, İstanbul
Pertevniyal Lisesi, Aksaray/Fatih, İstanbul

Abstract. Five completed, validity-checked SPH pilot simulations were run in a fixed DART impact scene to measure model sensitivity before production inference. At matrix cohesion $Y_0 = 10, 500$, and 5000 Pa, simulated momentum enhancement is $\beta = 3.7480, 2.6247$, and 1.8738 , respectively; the corresponding recorded ejecta masses are $1.964 \times 10^7, 3.649 \times 10^6$, and 7.516×10^5 kg. At 10 Pa, enabling the tested matrix-tension law reduces β to 1.0807 , while a 25 m impact offset gives 3.6650 and changes model ejecta mass by 18% . A three-point power law for $\beta - 1$ has slope -0.1791 and maximum relative residual 0.1198 ; its scene-conditioned interpolation gives 20.1 Pa at the mass-adjusted observed β . That interpolation is not an interior-strength measurement or posterior. We register a 96 -point coarse design and 12 nested fine reruns, and identify unresolved numerical and observation-definition checks that prevent a defensible production posterior or numerical Hera forecast at present.

1 Introduction

The DART experiment measured a change in Dimorphos’s orbit, constraining the momentum transferred to the body [1, 2]. Translating that response into interior structure requires a forward model whose material laws, target geometry, resolution, and observation operators are controlled. We use DART-RIFT to study three interior coordinates: block distension α_b , matrix cohesion Y_0 , and block volume fraction f . A match by one simulated scene is evidence about that scene, not an identified three-parameter interior.

This paper reports the completed pilot measurements first. It then specifies the production design and claim checks that were fixed before reading the production pool. The pilot runs themselves predate this paper and are not presented as blind predictions.

2 Simulation and comparison

2.1 Fixed scene and controlled changes

The reference target is a triaxial ellipsoid with semiaxes $(88.5, 87, 58)$ m and calibrated mass 4.2980×10^9 kg. The three-sphere impactor has mass 579.4 kg, speed 6144.9 m s⁻¹, and nominal incidence 17° from the surface normal. The pilot cohesion comparison fixes $(\alpha_b, f) = (1.15, 0.25)$, the random seed 20260906 , 7 m particle spacing, a 1.0 s transition time, and a 600 s endpoint [pilot protocol]. DO1 and DO2 change only Y_0 from the DY2 reference. DC enables one selected matrix-tension law at 10 Pa; DN moves the impact point 25 m

Table 1. Completed pilot runs at 600 s. Conditions other than the listed change are held fixed.

Run	Change	β	M_{ej} (kg)
DY2	$Y_0 = 10$ Pa	3.7480	1.964×10^7
DO1	$Y_0 = 500$ Pa	2.6247	3.649×10^6
DO2	$Y_0 = 5000$ Pa	1.8738	7.516×10^5
DC	Tension on	1.0807	2.777×10^2
DN	Offset 25 m	3.6650	2.321×10^7

at 10 Pa. The production model keeps that tension law disabled [model decision].

All four new branches completed and passed the repository’s validity checks; the reference DY2 also passed. Table 1 reports their recorded β and ejecta mass. The run identifiers, code commits, and file hashes are in the [run source record]. We use *model ejecta mass* for the simulation quantity; it is not automatically the same as an observed plume mass.

The four new branches have energy deviations between -0.79% and -0.80% , mass-consistency deviations between -0.05% and -0.06% , and momentum-ledger residuals no greater than 9×10^{-15} in the recorded checks [validation record]. These checks support run validity; they do not establish resolution convergence.

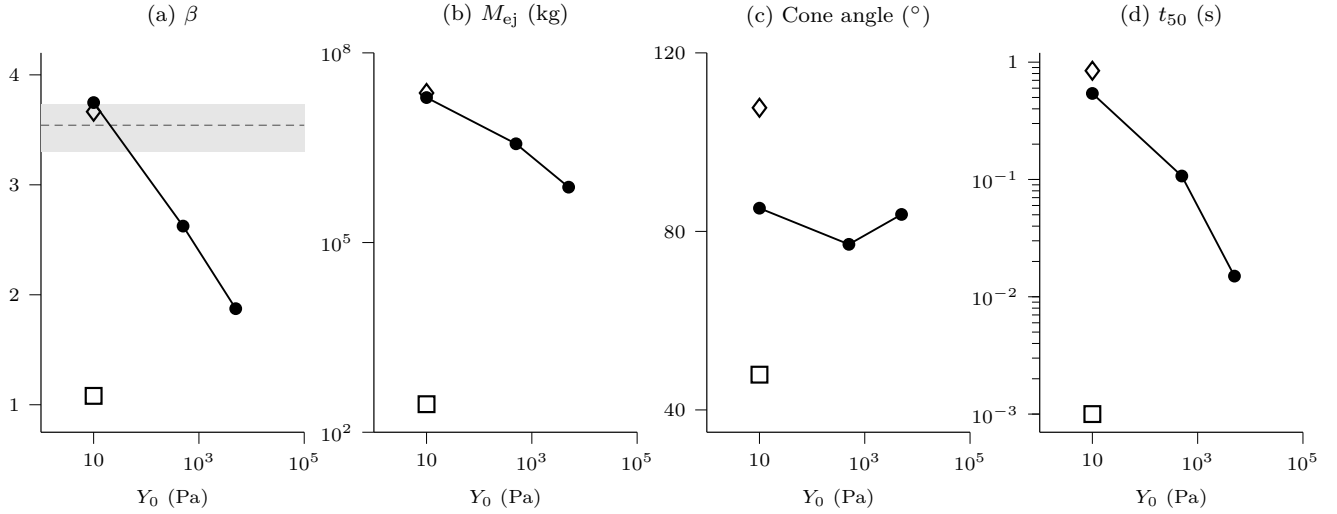


Figure 1. Four-observable atlas for the five completed pilot runs. Filled circles and connecting segments show only the three cohesion runs (DY2, DO1, DO2); the segments guide the eye and are not an inferred response curve. The open square is DC (matrix tension on) and the open diamond is DN (25 m impact offset); both retain $Y_0 = 10$ Pa but are not part of the cohesion sweep. Cone opening and median ejecta time t_{50} are model diagnostics. The gray band in panel (a) is the mass-adjusted DART 1σ interval from Equation 1; its dashed line is the central value. Panel (b) has no observational band because the model ejecta statistic is not the LICIAcube plume estimate.

2.2 Observational target

The mass-adjusted momentum target used for the production scene is

$$\beta_{\text{obs}}(M_{\text{scene}}) = 3.5418^{+0.188}_{-0.247} \quad (M_{\text{scene}} = 4.2980 \cdot 10^9 \text{ kg}). \quad (1)$$

The asymmetric interval is retained when constructing the likelihood [mass adjustment]. It is shown in Figure 1 only as an observational reference; it is not a posterior interval.

3 Measured pilot results

3.1 Cohesion sweep

Across the tested 10–5000 Pa span, β falls from 3.7480 to 1.8738 while model ejecta mass falls by a factor of 26.1 (Figure 1). The predeclared three-point fit $\beta - 1 = CY_0^p$ gives $p = -0.1791$ and a maximum relative residual of 0.1198, below the fixed 0.15 fit-quality threshold [DO protocol]. At the central target in Equation 1, this fit interpolates to $Y_0 \approx 20.1$ Pa. The value is conditional on the fixed scene, α_b , f , material law, and fit form; it is not a posterior estimate of Dimorphos’s strength. The older production cohesion domain $[10^3, 10^7]$ Pa does not contain this conditional interpolation, motivating the prospectively selected $[1, 10^5]$ Pa domain.

3.2 Matrix tension

At $Y_0 = 10$ Pa, enabling the tested tension law changes β from 3.7480 to 1.0807. The contrast in excess momentum $b = \beta - 1$ is

$$\frac{|b_{\text{DY2}} - b_{\text{DC}}|}{b_{\text{DY2}}} = \frac{|2.7480 - 0.0807|}{2.7480} = 0.9706. \quad (2)$$

Recorded model ejecta mass changes from 1.964×10^7 kg to 2.777×10^2 kg, a factor of about 7.1×10^4 . This is a strong sensitivity to one constitutive-model choice. Because DC is one toggle at one parameter location, it does not establish the real body’s tensile strength or rule out all alternative tension laws. The production model choice remains explicit, and the DC contrast is reported outside the small-discrepancy denominator [locked pilot judgments].

3.3 Impact-site offset

Moving the impact point 25 m changes β from 3.7480 to 3.6650 (2.2%), but changes model ejecta mass from 1.964×10^7 to 2.321×10^7 kg (18%). The recorded cone opening changes from 85.2° to 107.7° , and median ejecta time t_{50} from 0.541 to 0.845 s. A near-stable β therefore coexists with a materially different ejecta field in this controlled pair. The corresponding impact-location discrepancy on b is 0.03019; it exceeds the preregistered *negligible* boundary of 0.03000 by 0.00019. We retain the registered classification rather than moving that boundary after seeing the result [KAYIT-076]. Figure 3 compares the four paired responses on a common dimensionless scale and places the five runs in joint β –model-ejecta-mass space.

4 Registered production inference

4.1 Design and validation

The production coordinates are $\theta = (\alpha_b, Y_0, f)$, with $\alpha_b \in [1.00, 1.30]$, $Y_0 \in [1, 10^5]$ Pa, and $f \in [0.05, 0.50]$. The design has 96 Latin-hypercube coarse points at 7 m spac-

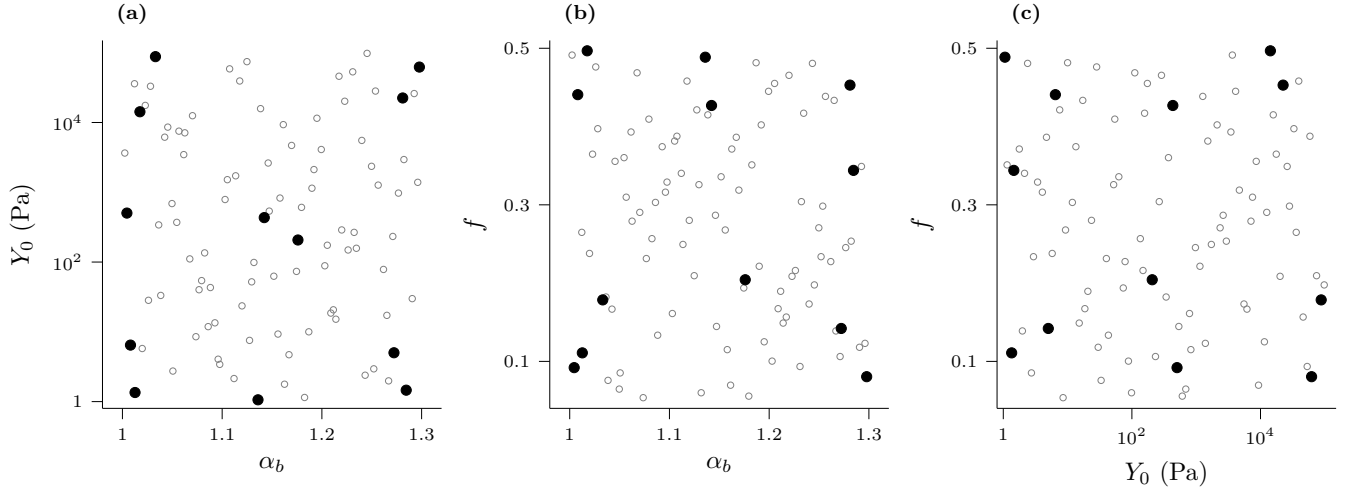


Figure 2. Locked design before production-pool inspection. Open circles are the 84 coarse-only points; filled circles are the 12 nested fine locations. The three projections show coverage of (α_b, Y_0, f) without suggesting measured responses or an inferred interior.

ing and 12 predetermined nested reruns at 3.5 m spacing [nested-design decision]. Figure 2 shows coverage; it is a design plot, not a fitted posterior. The 12 indices are $\{11, 20, 33, 44, 45, 57, 58, 62, 74, 78, 87, 92\}$. Every readable run must be valid, reach 600 s, and meet the 5% mass-consistency limit. A loss above 20% invalidates the pool and triggers its repetition [pool protocol].

The registered surrogate is a Gaussian process validated in four groups by θ , with predictive variance enlarged when held-out residuals require it. The residual-correlation matrix R is measured from those residuals and compared with $R = I$. The alternative resolution paths are a coarse surrogate plus measured discrepancy (SIGMA), and an autoregressive nested coarse/fine surrogate (CD). CD leads only if the nested design is intact, the fitted coupling is interior, and held-out fine error is lower than for SIGMA. Both results are disclosed if they differ materially [posterior definition]. For eligible observation vector z and surrogate mean $g(\theta)$, the registered heteroskedastic likelihood is

$$\mathcal{L}(\theta) \propto |S(\theta)|^{-1/2} \exp\left[-\frac{1}{2}(g - z)^T S(\theta)^{-1}(g - z)\right]. \quad (3)$$

The covariance $S(\theta)$ combines predictive, observation, and admitted model uncertainty with the measured residual correlation. The determinant term is retained; no production posterior is evaluated in this report.

4.2 Observation mapping and model error

A LICIACube analysis reports an extrapolated plume mass $(1.6 \pm 0.3) \times 10^7$ kg as a plausible upper limit for mass within the camera field of view [3]. The model quantity in Table 1 is the reported ejecta statistic at 600 s; the run record also tracks a separate escaping-mass diagnostic. Neither is definitionally identical to the observed plume mass. A quantitative ejecta-mass likelihood is conditional on an observation operator that reconciles them. The present model β is projected along impactor momentum, whereas the orbital observation concerns

Dimorphos’s orbital velocity direction. Both projections must be computed with verified orbital constants before physical posterior interpretation [post-processing protocol].

The predeclared relative discrepancy terms on $b = \beta - 1$ are 0.013 (realisation), 0.004 (far-field resolution), 0.053 (near-field resolution), 0.016 (late-time plateau), 0.009 (target shape), and 0.030 (impact location). The near-field value is documented but its executable constant is still nan; the code rejects that configuration. We therefore do not quote a complete operational denominator. The 0.134 impactor-geometry and 0.971 matrix-tension contrasts are alternative-model sensitivities, not Gaussian noise terms [discrepancy protocol].

4.3 Checks before a posterior claim

Simulation-based calibration uses at least 200 replicates with the actual surrogate and error model; a biased rank distribution blocks posterior publication [4]. Local Fisher information is evaluated at no fewer than five parameter locations. Eigenvalues above 36 count learned directions, and the smallest count over the registered correlation sensitivities governs the claim [identifiability protocol]. Only two quantitative observation groups are currently locked, so local rank cannot exceed two. The third candidate, ejecta-speed distribution, has no locked DART value and uncertainty.

A shift of more than 0.50 dex in the Y_0 mode under the declared prior alternatives prevents a single Y_0 estimate. Independent history matching excludes points with implausibility $I > 3$ [5]. The preregistered overlap and volume gates decide whether a posterior, a broader matching region, or neither is publishable. These gates are not adjusted after inspection of the production pool. If less than 90% of the 68% posterior mass lies in the compatible region, both inferential regions are withheld.

5 Discussion and conclusions

The pilot set supports three controlled findings. **First**, simulated momentum enhancement and ejecta mass decrease across the tested cohesion values in one fixed DART scene. **Second**, the selected matrix-tension toggle changes excess momentum by 97.1% and reduces the model-reported ejecta statistic by about 7.1×10^4 in that scene. **Third**, a 25 m impact offset changes β by only 2.2% while changing ejecta mass, opening angle, and median ejecta time more. These are model sensitivities, not a reconstructed interior.

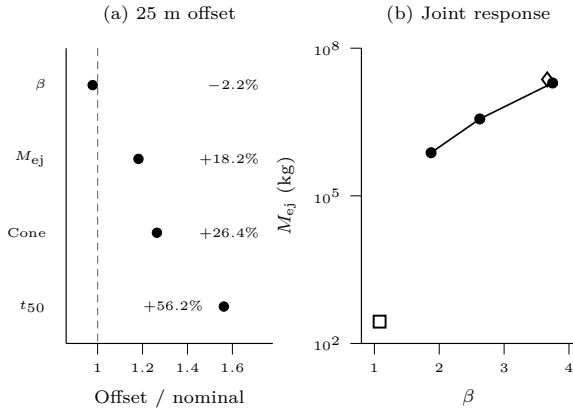


Figure 3. Two views of pilot sensitivity. (a) Each marker is the 25 m offset result divided by the nominal DY2 result; the dashed line is unity. (b) The joint β –model–ejecta response of all five runs: filled circles connected only along the cohesion sweep, open square for DC, and open diamond for DN. Panel (b) is a model-response plot, not a fitted relation or an observed ejecta-mass comparison.

The principal limits are structural. The three-point cohesion fit is conditional on the pilot scene. The near-field discrepancy has not been connected in code. The plume and model–ejecta definitions differ, as do the observed and simulated momentum-projection axes. Crater depth at the present 600 s endpoint is invalid or physically mismatched to a final settled crater in the pilot diagnostics [crater diagnosis]. A single earlier resolution step measured 5.34% on $\beta - 1$ in a different scene; it is not a convergence proof [resolution record].

The Hera protocol defines quantile and hash-sealing procedures, but the required posterior and validated crater/shape post-processing are not yet available [Hera protocol]. No numerical Hera forecast or production posterior is reported here.

Data and protocol provenance

The pilot values and locked decisions are archived at commit a35dab4. Figures 1 and 3 use the completed run diagnostics in Table 1 and the [locked pilot record]; Figure 2 uses the locked design coordinates. Later analyses must retain these dated decision rules. The article text and original figures are © 2026 the authors, licensed under CC BY 4.0; the Ethosoft mark, ORCID iD icon, and GitHub mark are excluded. Repository software remains under the MIT license.

Prospective Hera comparison

Hera will measure Dimorphos’s mass, map its crater and surface, and investigate its interior [6]. Table 2 separates pilot evidence from future checks.

Table 2. Pilot claims and possible Hera checks. No numerical forecast is sealed yet.

Question	Evidence and independent check
Cohesion	β falls $3.7480 \rightarrow 1.8738$ and M_{ej} falls $26.1 \times$ over 10–5000 Pa. Hera’s mass measurement refines β_{obs} ; crater and shape can test later predictions. The 20.1 Pa fit is scene-conditional.
Matrix tension	The tested toggle changes β $3.7480 \rightarrow 1.0807$. Hera’s interior and surface data can constrain competing models; one toggle cannot prove zero real tensile strength.
Impact site	A 25 m offset changes β by -2.2% and M_{ej} by $+18.2\%$. Hera can map the crater and site, but cannot remeasure the past ejecta time t_{50} .

Two planned DOI records

DOI 1 (planned; this paper). Five validated pilot runs, quantitative sensitivities, and registered inference rules.

DOI 2 (planned). The detailed production posterior or matching region, uncertainty analysis, and numerical Hera predictions, if the claim gates pass. Neither DOI is assigned here. A discriminating Hera comparison requires time-stamped predictions and judgment rules before the relevant measurements are inspected.

References

- [1] T. R. Daly et al., Successful kinetic impact into an asteroid for planetary defence, *Nature* **616**, 443–447 (2023). doi:10.1038/s41586-023-05810-5.
- [2] A. F. Cheng et al., Momentum transfer from the DART mission kinetic impact on asteroid Dimorphos, *Nature* **616**, 457–460 (2023). doi:10.1038/s41586-023-05878-z.
- [3] R. Lolachi et al., Ejecta Mass Estimates from the DART Impact Plume Inferred from LICIACube Images, *Planetary Science Journal* **6**, 199 (2025). doi:10.3847/PSJ/adec6b.
- [4] S. Talts et al., Validating Bayesian Inference with Simulation-Based Calibration, arXiv:1804.06788 (2018).
- [5] I. Vernon, M. Goldstein, and R. G. Bower, Galaxy Formation: Bayesian History Matching for the Observable Universe, *Statistical Science* **29**, 81–90 (2014). doi:10.1214/12-STS412.
- [6] European Space Agency, *Hera’s Science* (accessed 10 October 2026). esa.int.